

Nock API Reference

The Nock API gives partners programmatic access to the Nock agents on Robinhood Chain — swaps, yield, tokenized stocks, stock-backed lending, and perps. Read live market data, or build a transaction and sign it yourself. Nock never holds your keys.

Base URL `https://app.nockfi.io/api/v1`

Chain `Robinhood Chain (4663)`

Auth `API key`

Custody model. Every endpoint is read-only on our side. Endpoints that create an action return an **unsigned transaction** (or an ordered list of them); your application signs and broadcasts. Keys, approvals, and enforcement stay entirely with you.

AUTHENTICATION

Every request must carry your API key. Requests without a valid key return `401` — no endpoint is public.

Pass your key in the `x-api-key` header (or `Authorization: Bearer <key>`). We issue one key per partner and store only its hash. Requests are rate-limited per key; contact us to raise limits.

REQUEST

CURL

```
curl https://app.nockfi.io/api/v1/yield/markets \  
-H "x-api-key: npk_live_your_key_here"
```

SWAP AGENT

GET /swap/quote EXECUTABLE

Returns the best-price route between two tokens on Robinhood Chain (routed via 0x), including an unsigned transaction to execute the swap. Every quote carries a `verified` flag — `false` means at least one token is an unvetted/impersonator contract.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
<code>fromToken</code>	string	required	Token to sell — a symbol (<code>USDG</code>) or a contract address.
<code>toToken</code>	string	required	Token to buy — symbol or contract address.
<code>amount</code>	string	required	Human amount of <code>fromToken</code> , e.g. <code>"100"</code> .
<code>taker</code>	string	optional	Wallet that will sign the transaction. Include it to get an executable <code>transaction</code> .

REQUEST

GET

CURL

```
curl ".../v1/swap/quote?fromToken=USDG\
&toToken=NOCK&amount=100\
&taker=0x6690...Aab1" \
-H "x-api-key: npk_live_..."
```

RESPONSE 200

JSON

```
{
  "fromSymbol": "USDG",
  "toSymbol": "NOCK",
  "toAmount": "5404.03",
  "exchangeRate": "1 USDG = 540.4 NOCK",
  "liquidityAvailable": true,
  "verified": true,
  "sellTokenAddress": "0x5fc5...d168",
  "sellTokenDecimals": 6,
  "transaction": {
    "to": "0x0000...2734",
    "data": "0x...",
    "value": "0", "gas": "250000"
  }
}
```

GET /yield/markets

Lists the available lending / vault markets with live APY, TVL, and risk label (Morpho, Steakhouse).

QUERY PARAMETERS

None.

RESPONSE

JSON

200

```
{ "markets": [
  { "name": "Steakhouse USDG", "assetSymbol": "USDG",
    "apy": 4.2, "totalAssetsUsd": 1250000,
    "risk": "...", "protocol": "Morpho", "address": "0x..." }
] }
```

GET /yield/positions

A wallet's live supply positions with current APY.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
address	string	required	Wallet address to read positions for.

RESPONSE

JSON

200

```
{ "positions": [ { "market": "...", "suppliedUsd": 100, "apyPct": 4.2 } ] }
```

GET `/yield/deposit`**EXECUTABLE**

"Put money to work." Builds the unsigned deposit transaction into the yield vault. Sign and broadcast it to execute. Returns a `422` with a plain-English reason if the vault can't accept the deposit.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
<code>address</code>	string	required	Depositing wallet.
<code>amount</code>	string	required	USDG amount to deposit, e.g. <code>"100"</code> .

RESPONSE

JSON

200

```
{
  "transaction": { "to": "0x...vault", "data": "0x...", "value": "0", "gas": "300000" },
  "vaultSymbol": "...", "sharesPreview": "...", "amount": "100"
}
```

TOKENIZED STOCKS**GET** `/stocks`

The verified tokenized-equity universe (AAPL, TSLA, NVDA, SPY, ...), each checked against Robinhood's official issuer.

QUERY PARAMETERS

None.

RESPONSE

JSON

200

```
{ "stocks": [ { "symbol": "TSLA", "name": "Tesla", "address": "0x...", "priceUsd": 385
```

GET /stocks/quote EXECUTABLE

Buy or sell a tokenized stock (routed through Uniswap). Returns the price and the unsigned transaction — sign it to trade.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
<code>symbol</code>	string	required	Stock symbol, e.g. <code>TSLA</code> .
<code>direction</code>	string	required	<code>buy</code> or <code>sell</code> .
<code>amount</code>	string	required	USDG for a buy; stock units for a sell.

RESPONSE

JSON

200

```
{
  "fromSymbol": "USDG", "toSymbol": "TSLA",
  "toAmount": "0.012962", "verified": true, "routeVia": "uniswap-v4",
  "transaction": { "to": "0x...", "data": "0x...", "value": "0", "gas": "..." }
}
```

LENDING — BORROW AGAINST STOCKS

GET /collateral/markets

Which tokenized stocks can back a USDG loan, with live borrow APY, max LTV (LLTV), and available liquidity.

QUERY PARAMETERS

None.

GET /collateral/positions

A wallet's open loans: debt owed, collateral value, and liquidation price.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
address	string	required	Wallet address.

GET /collateral/borrow EXECUTABLE

Post tokenized-stock collateral and borrow USDG (e.g. post TSLA, borrow USDG). Returns the ordered step transactions plus any required ERC-20 approval, and the resulting LTV, borrow APY, and liquidation price so you can gate on risk. Over-leveraged requests are rejected with the safe maximum.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
address	string	required	Borrowing wallet.
stock	string	required	Collateral stock symbol, e.g. <code>TSLA</code> .
borrowUsd	string	required	USDG amount to borrow.
collateralAmount	string	optional	Stock units to post. Defaults to the wallet's full balance.

RESPONSE

JSON

200

```
{
  "kind": "stock-borrow", "stockSymbol": "TSLA",
  "approval": { "tokenAddress": "0x...", "amountRaw": "...", "spender": "0x..." },
  "steps": [ { "to": "0x...", "data": "0x...", "value": "0" } ], // sign in order
  "usdgAmount": "...", "borrowApyPct": 6.1,
  "ltvUtilizationAfterPct": 42.0, "liquidationPriceUsdAfter": ...
}
```

GET /perps/markets

Live perpetual-futures market data on Robinhood Chain (via Lighter): mark price, hourly funding, open interest, 24h volume, and max leverage. Read-only — live order execution is jurisdiction-gated and not exposed via the API.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
<code>symbol</code>	string	optional	Filter to one market, e.g. <code>ETH</code> . Omit for the top markets by volume.

RESPONSE

JSON

200

```
{ "markets": [  
  { "asset": "ETH", "markPrice": 1831.61, "fundingRatePctHourly": 0.0096,  
    "openInterest": 49610, "dailyVolumeUsd": 286985432, "maxLeverage": 20 }  
]
```

VAULT & PORTFOLIO

GET /vault/guardrails

The safety guardrails a wallet has set — currently the per-transaction USD spend limit that gates any swap or deposit. Mirror it in your own pre-trade checks.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
<code>address</code>	string	required	Wallet address.

RESPONSE

JSON

200

```
{ "address": "0x...", "maxUsdPerTransaction": 500 } // null = no limit set
```

GET /portfolio

Everything a wallet holds in one call — token and stock-token balances with live USD values and a portfolio total.

QUERY PARAMETERS

PARAMETER	TYPE	REQUIRED	DESCRIPTION
address	string	required	Wallet address.

RESPONSE

JSON

200

```
{
  "address": "0x...", "totalPortfolioUsd": 17.63,
  "balances": [
    { "symbol": "USDG", "amount": "15.13", "usdValue": 15.13 },
    { "symbol": "NOCK", "name": "Nock Finance", "amount": "...", "usdValue": ... }
  ]
}
```

Nock — AI agent concierge on Robinhood Chain (4663). One API key per partner; contact the Nock team for access and to raise rate limits. · app.nockfi.io